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Class-Controlled Histopathology Image Synthesis for Robust Tumor-Infiltrating Lymphocyte Segmentation

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ABSTRACT

Accurate assessment of tumor-infiltrating lymphocytes (TILs) in melanoma histopathology is essential for predicting immunotherapy response and improving patient outcomes. However, current automated segmentation methods are hindered by limited datasets and severe class imbalance. We propose a novel dual-generator adversarial framework that transforms synthetic histopathology data generation by decomposing the process into two sequential, specialized tasks: controllable mask generation with user-defined class distributions, followed by high-fidelity histopathology image synthesis. This approach enables precise dataset augmentation with any desired number of tissue classes per image, directly addressing the scarcity of balanced training data. Using the PUMA Grand Challenge dataset, we generated two complementary datasets and evaluated them with a custom U-Net architecture integrating a powerful MedSAM encoder and a decoder optimized for fine-grained tissue segmentation. The proposed framework excels in producing photorealistic histopathology images while preserving precise control over tissue class distributions and spatial relationships. The proposed architecture achieved Dice (0.92) and accuracy (0.97) on both the PUMA dataset and a newly generated fourth-class-per-image dataset, marking a significant advance in melanoma tissue segmentation. This scalable framework establishes a new paradigm for computational pathology and enhancing decision-making in melanoma management.

Keywords: Melanoma, Histopathology, GANs, Synthetic Data Generation, MedSAM, Semantic Segmentation

1. INTRODUCTION

Accurate characterization of tumor-infiltrating lymphocytes (TILs) is essential for evaluating immune response and improving treatment decisions in melanoma. TIL density and spatial distribution have been shown to correlate strongly with patient prognosis and response to immunotherapy (1; 2). In routine clinical practice, however, TIL assessment is typically performed manually on hematoxylin and eosin (H&E) slides, a process that is inherently subjective and limited by interobserver variability (3; 4). This motivates the development of robust automated segmentation systems capable of capturing fine-grained tissue morphology and cellular interactions within the tumor microenvironment. Despite recent progress in computational pathology, segmentation of melanoma tissues remains challenging due to limited availability of curated datasets, severe class imbalance, and the morphological diversity of advanced melanoma lesions. Existing public datasets such as PanNuke (5) and MoNuSAC (6) provide valuable annotations but are not melanoma-specific and fail to capture the full heterogeneity of tumor architecture (7). The Panoptic Segmentation of nUclei and tissue in advanced MelanomA (PUMA) dataset addresses this gap, offering high-quality annotations of melanoma tissues (8; 9). However, its modest size and uneven representation of tissue classes restrict the performance and generalization of deep segmentation models. These limitations highlight the need for scalable synthetic data generation methods capable of producing realistic and diverse histopathology samples with controllable tissue composition.

Generative models, particularly Generative Adversarial Networks (GANs), have demonstrated strong potential for data augmentation in medical imaging. Prior work has shown that GAN-generated histopathology patches

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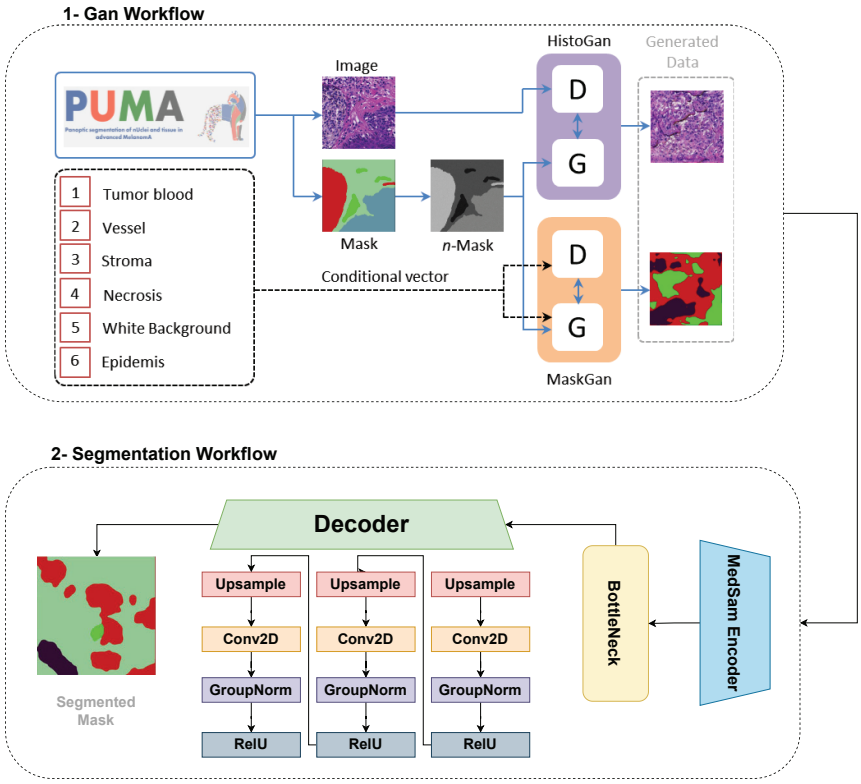

Figure 1. Overall workflow of the proposed framework showing the two-stage process: (1) GAN-based synthetic data generation and (2) MedSAM-based tissue segmentation.

can enhance model robustness by introducing morphological diversity and compensating for rare classes (10). However, most existing approaches focus on global image translation or stain-style adaptation rather than generating structurally valid, class-controlled tissue masks that explicitly encode spatial arrangement and class composition. Such controllability is crucial for augmenting underrepresented tissue types and improving segmentation of complex melanoma regions.

To address these challenges, we introduce a class-controllable dual-generator framework that decomposes synthetic data generation into two stages: (1) a Mask Generator that produces semantic masks with user-specified tissue classes, and (2) an Image Generator that synthesizes realistic melanoma histopathology images conditioned on these masks. This design enables precise control over class distributions, allowing the generation of balanced datasets that better capture the diversity of melanoma morphology. Furthermore, we integrate these synthetic datasets with a MedSAM-based segmentation model (11) to evaluate the impact of structured augmentation on tissue-level prediction. We additionally introduce a strengthened mask generation experiment in which the Mask GAN simultaneously produces from two to four distinct tissue masks per input noise vector. This enhanced formulation significantly improved spatial diversity and class co-occurrence modeling, resulting in the highest segmentation performance observed in our experiments when training the MedSAM-based segmentation architecture. The contributions of this study are: (i) a dual-stage GAN framework for controllable melanoma histopathology synthesis, (ii) a multi-mask generation strategy that improves structural diversity, and (iii) a comprehensive evaluation showing substantial gains in tissue segmentation performance. These findings highlight the potential of class-controlled synthetic data to strengthen automated TIL and melanoma tissue analysis, supporting scalable computational pathology workflows and future precision oncology applications.

2. METHODS

This study introduces a two-stage framework that integrates controllable synthetic data generation with a MedSAM-based segmentation model as illustrated in Figure 1. The framework is designed to alleviate the challenges of limited annotated data, class imbalance, and morphological variability in melanoma histopathology. The workflow begins with a dual-GAN pipeline for generating masks and realistic histopathology images, followed by a segmentation model that leverages both original and synthetic datasets.

2.1 Dataset

All experiments were conducted using the PUMA dataset, which comprises 310 regions of interest (ROIs) derived from H&E-stained whole-slide images of primary and metastatic melanoma. Each ROI was digitized at 40× magnification (0.23 μm/px) with a fixed resolution of 1024×1024 pixels. Expert pathologist annotations delineate six tissue categories: tumor, stroma, epidermis, necrosis, blood vessels, and background. Primary melanoma ROIs exhibit a relatively balanced proportion of tumor and stromal regions, whereas metastatic ROIs contain approximately 70% tumor tissue, resulting in pronounced class imbalance. The dataset is evenly divided into 155 primary and 155 metastatic ROIs, each provided in GeoJSON format. These annotations are used as ground truth for training the generative models and for evaluating the downstream segmentation framework.

2.2 GANs Workflow

The synthetic data pipeline consists of two conditional GANs arranged sequentially. The first network, referred to as the Mask GAN, generates 256×256 segmentation masks that contain any subset of the six PUMA tissue classes. To make this process intuitive, each tissue class is associated with a dedicated numeric interval that defines the range from which its noise values are sampled. Considering that the image $I(x, y)$ is segmented into tissue regions $\{\Phi_{t_1}, \Phi_{t_2}, \ldots, \Phi_{t_n}\}$, and $T = \{t_1, t_2, \ldots, t_n\}$ is the pool of n tissue labels in ascending order, we first define normalized label values

$$t'_i = \frac{t_i - \min(T)}{\max(T) - \min(T)}, \quad i = 1, \ldots, n. \tag{1}$$

We then define boundary values $B = \{b_1, b_2, \ldots, b_{n-1}\}$ as

$$b_i = t'_i + \frac{t'_{i+1} - t'_i}{2}, \quad i = 1, \ldots, n-1. \tag{2}$$

The perturbed mask $\tilde{I}(x, y)$ is then defined by

$$\epsilon_{(x,y)} \sim \mathcal{U}(b_i, b_{i+1}), \quad \forall (x, y) \in \Phi_{t_i}, \tag{3}$$

$$\tilde{I}(x, y) = \epsilon_{(x,y)}, \tag{4}$$

where $\mathcal{U}(a, b)$ denotes the uniform distribution over $[a, b)$. When these values are arranged over the spatial grid, the result is a grayscale image whose local texture corresponds to the noise distribution of the underlying class. Thus, the mask appears as contiguous regions with different granular noise patterns, as illustrated in Figure 2, where darker regions correspond to classes assigned lower labels (e.g., blood vessels), and brighter regions correspond to classes with higher labels (e.g., background). This provides the GAN with a rich, class-specific spatial signal while preserving the exact geometry of the anatomical structures.

The second network, referred to as the Histopathology Image GAN, transforms the generated masks into realistic H&E-style images. It adopts a U-Net encoder–decoder structure with skip connections that preserve spatial detail across the synthesis process. The input mask is first perturbed with Gaussian noise to simulate realistic variability before being processed by the encoder. The decoder reconstructs a three-channel RGB image through transposed convolutions, with dropout applied in early layers to improve generalization. The discriminator employs a PatchGAN design, evaluating image realism at the patch level to enforce high-frequency texture consistency. Training is guided by a combination of adversarial loss and an L1 reconstruction term, encouraging both perceptual realism and structural faithfulness to the underlying mask. By sampling different noise and condition vectors, this dual-GAN pipeline yields large-scale synthetic datasets with precisely controlled tissue compositions, including the extended four-class mask configuration used in this study.

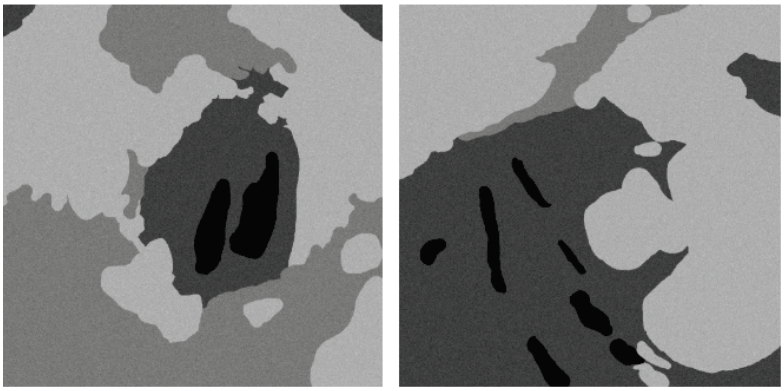

Figure 2. Randomly sampled 2D masks (Mask 1 and Mask 2) generated from the 1D class distribution vector.

2.3 Segmentation Architecture

To assess the utility of the synthetic datasets, we adopt a segmentation architecture built around the MedSAM Vision Transformer encoder (11). The encoder operates in a frozen state to retain its pretrained medical imaging priors and to reduce the risk of overfitting on the limited original dataset. Input images are first tokenized into patch embeddings via a convolutional projection layer. These embeddings pass through a sequence of transformer blocks that model long-range dependencies and global contextual relationships across the histopathology image.

A neck module transforms the transformer output into a feature map suitable for dense prediction by applying a 1×1 convolution to reduce the channel dimension, layer normalization to stabilize the distribution of activations, and a 3×3 convolution to inject local spatial structure. The custom decoder then reconstructs pixel-level segmentation maps through four stages of bilinear upsampling followed by convolutional refinement. Group normalization and ReLU activations are used throughout the decoder to ensure stable gradient flow and consistent feature scaling. Channel dimensionality is progressively reduced until it reaches six channels, corresponding to the six PUMA tissue classes. The final segmentation map is produced through a per-pixel softmax function. Training follows a 5-fold cross-validation setup, optimizing cross-entropy loss with the AdamW optimizer for five epochs per fold, leveraging both real and synthetically generated images depending on the dataset configuration.

This three-part methodology, dataset preparation, controllable dual-GAN synthesis, and MedSAM-based segmentation, forms a unified framework for generating and utilizing synthetic histopathology data to improve tissue-level classification in melanoma.

3. RESULTS

To assess the impact of class-controlled synthetic data on melanoma tissue segmentation, the CustomUNet model (MedSAM encoder with custom decoder) was evaluated on four dataset configurations using 5-fold cross-validation. Training was performed for five epochs per fold across the following datasets: the original PUMA dataset (205 images), a 2-class GAN-augmented dataset (1000 generated + 205 original), a 3-class GAN-augmented dataset (1205 total), and an extended 4-class GAN-augmented dataset (1205 total). Table 1 summarizes the averaged Dice and accuracy scores across all folds.

Table 1. Average segmentation performance across datasets (5 epochs, 5 folds).

Dataset Configuration	**Avg. Dice**	**Avg. Accuracy**
Original Data (205 images)	0.74	0.94
2-Class (1000G + 205OD)	0.90	0.97
3-Class (1000G + 205OD)	0.91	**0.98**
4-Class (1000G + 205OD)	**0.92**	0.97

The results demonstrate that the inclusion of controllable synthetic data notably enhances segmentation performance compared to training on the original PUMA dataset alone. The 2-class and 3-class augmented datasets

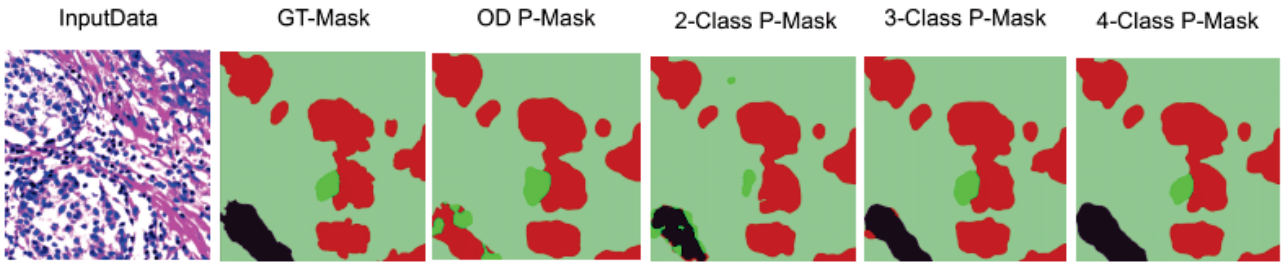

Figure 3. Comparison of segmentation outputs for different dataset configurations: Original Data, 2-Class GAN, 3-Class GAN, and 4-Class GAN. Each column shows the input H&E patch, the corresponding ground-truth mask (GT Mask), and the predicted segmentation (P-Mask). Improvements become more pronounced as the synthetic class diversity increases.

Figure 4. Average training and validation metrics across 5 folds for the 4-class GAN-augmented dataset. Curves include loss, Dice, and F1-score, showing stable convergence.

both produced significant improvements, particularly in Dice score, with the 3-class configuration achieving the highest accuracy (0.98). The newly introduced 4-class dataset yielded the highest Dice score overall (0.92), indicating that increasing the diversity and structural complexity of the synthetic masks further strengthened the model's ability to generalize across tissue boundaries.

Figure 3 provides a visual comparison of segmentation outputs across all four dataset configurations. For each configuration, the figure displays the original input image, the ground-truth (GT) tissue mask, and the predicted mask (P-Mask). As the number of synthetic classes increases from two to four, the predictions become progressively more consistent with the GT masks, especially in regions containing stroma–tumor boundaries and small tissue structures. The 4-class model exhibits the closest alignment with the ground-truth, demonstrating improved preservation of tissue morphology and reduced over-segmentation artifacts. To further analyze the effect of synthetic augmentation, Figure 4 plots the average training and validation metrics across the 5 folds for the 4-class dataset. The metrics include loss, Dice score, and F1-score. All metrics show stable and consistent convergence across folds, with validation curves closely following the training curves. The Dice and F1 curves plateau around 0.92–0.93 reflecting improved segmentation quality and reduced variability across folds. The loss plot confirms steady optimization without overfitting, supported by the tight coupling between training and validation trajectories. These trends highlight the effectiveness of the 4-class augmentation strategy in producing a well-balanced dataset that enhances both performance and training stability.

4. CONCLUSION

This work presents a controllable synthetic data generation framework that significantly enhances melanoma tissue segmentation through a combination of dual-GAN mask and image synthesis and a MedSAM-based segmentation architecture. By decomposing the synthesis problem into mask generation and histopathology image translation, the framework enables precise control over tissue class composition, allowing the creation of diverse and balanced training datasets tailored to segmentation needs. This design directly addresses the challenges of dataset scarcity and class imbalance inherent in melanoma histopathology.

Across extensive experiments using the PUMA dataset, synthetic augmentation consistently improved segmentation performance compared to training on real data alone. The introduction of the four-class synthetic mask configuration, in particular, provided the most diverse structural patterns and yielded the highest Dice score of 0.92. This demonstrates that increasing the compositional complexity of the synthetic data leads to more robust generalization, improved boundary delineation, and greater consistency across folds. The integration of MedSAM as a frozen encoder further strengthened the segmentation pipeline by leveraging its pretrained medical imaging priors, enabling efficient fine-tuning even with limited real data. The combination of class-controlled GAN synthesis and a strong encoder–decoder model establishes a practical and scalable solution for advancing tissue-level analysis in computational pathology. To conclude, this study highlights the potential of controllable synthetic histopathology generation as a powerful tool for improving segmentation performance in domains where annotated data remains scarce. By enabling flexible dataset construction and enhancing segmentation robustness, the proposed framework offers a promising direction for future research in melanoma analysis, TIL quantification, and broader digital pathology applications.

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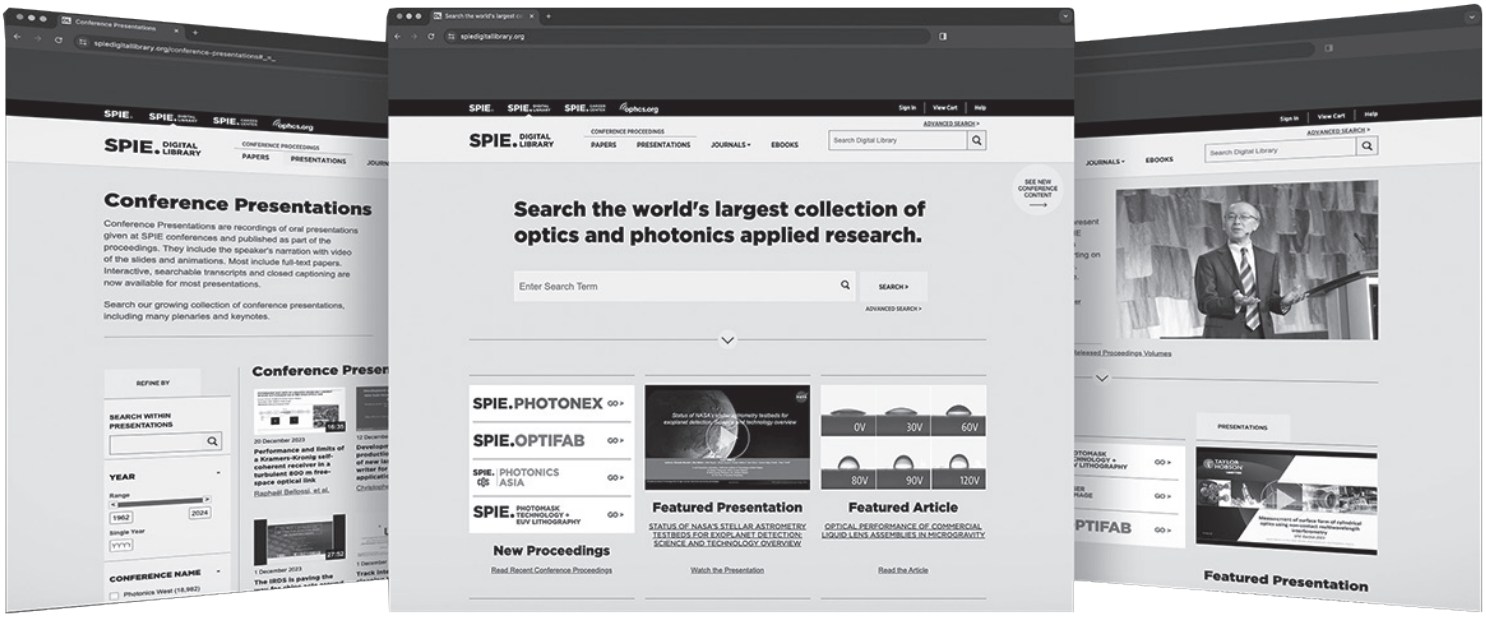

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